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The relationship between climate change and mental health: a systematic review of the association between eco-anxiety, psychological distress, and symptoms of major affective disorders

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Abstract

Background and objectives The adverse impacts of climate change on mental health is a burgeoning area, although findings are inconsistent. The emerging concept of eco-anxiety represents distress in relation to climate change and may be related to mental health. The aim of this study was to explore the relationship between eco-anxiety with validated mental health outcomes, specifically psychological distress and symptoms of major affective disorders.

Design Systematic review.

Methods EBSCO, ProQuest, and Web of Science databases were searched to February 2024 for studies of adult samples quantifying eco-anxiety (exposure, i.e. fear, worry or anxiety in relation to climate change) and symptoms of psychological distress and major affective disorders (outcomes), as assessed by validated measures.

Results Full text review of 83 studies was performed, and $k=35$ studies were included in the review ($N=45\,667$, 61% female, M_{age} 31.2 years). Consistently, eco-anxiety showed small to large positive correlations with mental health outcomes of psychological distress, depression symptoms, anxiety symptoms, and stress symptoms. However, results regarding post-traumatic stress disorder symptoms and pathological worry were mixed. Stronger associations were observed where eco-anxiety was operationalised as 'anxiety' rather than 'worry'.

Conclusions Findings underscore that eco-anxiety is related to psychological burden. Greater consideration of eco-anxiety in assessment and treatment is needed in clinical practice and further policy development is warranted at the intersection of climate and health to address the mental health challenges posed by climate change.

Keywords Eco-worry, Climate change distress, Eco-stress, Psychological disorders, Ecological worry, Wellbeing, Climate change anxiety, Mental health disorders

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Climate change represents a significant global threat that will have broad-ranging effects on human health [1]. While the physical health impacts are well established see [2], the *mental* health effects have only become a research focus in more recent years. Climate change is understood as impacting mental health through three pathways see [3, 4]: (a) direct exposure to climate-related extreme weather events (e.g., floods, wildfires, drought), (b) indirect exposure such as impacts resulting from extreme weather events (e.g., disruption of access to resources, forced relocations), and (c) via the *awareness* of climate change. This review focuses on the latter, exploring a vicarious pathway in which distress may arise through knowledge and awareness of environmental decline [5].

Reports of negative reactions to awareness of climate change have been reported since at least 2007 [6]. Over time, a plethora of terms has emerged describing a suite of eco-related emotions, with “eco-anxiety” most widely used to encompass negative emotions related to climate change and environmental degradation [7]. Eco-anxiety is defined by the American Psychological Association as “a chronic fear of environmental doom” (8, pg.68), encompassing feelings of worry, fear, and anxiety that arise from awareness of climate change [7, 8].

Investigations into anxiety about climate change have revealed that it is increasingly common for many people across the globe [9], with large-scale international studies identifying growing concern about climate change [10–12]. For example, recent multinational studies have suggested that between 49 and 59% of respondents were very or extremely worried about climate change [10, 11]. This increasing prevalence of eco-anxiety underscores the need to explore its relationship with mental health outcomes [9, 13]. While reports from clinical practice e.g., [14, 15] and mental healthcare professionals e.g., [16, 17] indicate that distress is occurring in relation to climate change, others argue that the threats of climate change are real and thus eco-anxiety is rational, rather than potentially pathological [18–20]. A review and synthesis on contemporary understandings of the extent to which anxiety around climate change can impact mental health is therefore urgent to guide policy makers and healthcare providers [21].

Recent systematic and scoping reviews have explored diverse concepts such as climate change perceptions or emotions [22, 23], reported only on the direct impacts of climate change [24–28], or have focused on adolescents [29]. Most critically, existing reviews are further limited by inclusion of studies that utilise non-validated assessments of mental health e.g., [22, 23, 29–31], limiting understanding of mental health outcomes and psychiatric burden associated with eco-anxiety and resulting in uncertainty on the potential vicarious impacts of climate

change on clinically-relevant distress. To date, there remains a lack of systematic synthesis of findings pertaining specifically to *validly* assessed mental health outcomes. Thus, the current review extends on past reviews by examining only studies utilising validated assessment of mental health outcomes. Synthesising literature for specific and validly assessed mental health outcomes is critical in order to guide and inform clinical practice. Therefore, the aim of this systematic review was to synthesise the relationship between eco-anxiety with mental health, specifically with validly assessed psychological distress and symptoms of major affective disorders.

Methods

The systematic literature review was pre-registered with the International Platform of Registered Systematic Reviews and Meta-Analysis Protocols (INPLASY202310025) and was conducted in accordance with the PRISMA guidelines [32].

Search strategy

Three databases were searched: EBSCO, ProQuest, and Web of Science until February 2024. Due to a lack of consensus on terminology and definition of eco-anxiety in the literature [7], we adopted a broad search strategy encompassing a variety of terms conceptually related to eco-anxiety (e.g., solastalg*, psychoterra*, ecoanxi*); see supplementary materials for full search string. Databases were searched by title and abstract, with two limiters applied: English language, and peer-reviewed. No time limits were applied. A hand search of the reference lists of included articles and other reviews e.g., [23, 30] was also performed.

Eligibility criteria

Full inclusion and exclusion criteria are presented in Table 1. As an emerging area of research, the population criterion was kept broad and there was no requirement for a comparator group. The focus was on adults, with age inclusion from 15 years and older, noting that most psychiatric illnesses have presented by late adolescence [33]. Given the varied conceptualisations in the literature of climate-related distress, the term “eco-anxiety” was defined as per Coffey et al. [7] as worried, anxious, or fearful responses (i.e. climate-change related worry, fear, and anxiety) to the *awareness* of climate change, rather than from the direct experience of extreme weather events, and as distinct from other eco-emotions such as eco-grief.

Eligible outcomes were any objective or self-reported measure of psychological distress or common major affective or related disorders and their core symptoms, assessed either concurrently or longitudinally. Due to the

Table 1 Criteria for inclusion/exclusion of studies

Parameter	Inclusion criteria	Exclusion criteria
Population (P)	Studies that were performed in adult (15+) human populations in any region and derived from the general population, community, or mental health setting	Studies reporting on child or adolescent populations (below age 15) or were derived from persons who were institutionalised, incarcerated, or unable to provide informed consent
Exposure (E)	Studies reporting on eco-anxiety – defined as fear, worry, or anxiety that arises from awareness of climate change, quantified by any measure, including single-item questions	Studies reporting on distress that arises from direct personal experience of extreme weather events (e.g., flood, bushfire, drought) or climatic conditions (e.g., ambient temperatures, seasonal changes) or related environmental impacts (e.g., changes in crop production, changes in marine life). Studies reporting on distress that arises from personal experience of psychosocial impacts of extreme weather events (e.g. loss of income or housing due to landslide). Exposures not encapsulated in ‘eco-anxiety’ such as eco-guilt; or attitudes towards or feelings about climate change or the environment
Comparator (C)	Studies do not need to refer to a comparison/control group to be included	N/A
Outcome (O)	Studies that report on mental health outcomes, encompassing major affective disorders (i.e., PTSD, depression, anxiety) and their symptoms (i.e., worry, distress), and behaviour (i.e., self-harm, suicide), as assessed by objective or validated self-report measures	Studies that do not report psychological distress, psychological disorders or mental health symptoms, studies that do not use validated assessment of mental health outcomes
Study design	Quantitative studies of any design with sample size ≥ 30 persons	Qualitative studies, case series < 30 persons, case-studies, dissertations, conference abstracts, editorials, letters, personal reflections, literature reviews, narrative summaries

limited literature in the field, quantitative studies of any design were eligible if reported as a full article. Ineligible research included qualitative studies, case-series < 30 persons or case-studies, or articles not reporting unique data such as reviews, editorials, and commentaries. Studies not published in English were also ineligible.

Screening and data extraction

Two of four reviewers (RR, KR, JT, KF) independently conducted eligibility assessment and disagreements were resolved by discussion with an additional reviewer (AL), and then verified by a final reviewer (SC). Study data were extracted by two authors (RR, KF) and verified by three independent raters (KR, JT, SC) according to a standard template which included identifying information (author, year, country), study characteristics (design, sample size), population characteristics (mean age, gender), exposure and outcome assessment tools, and findings relevant to this review's objectives. As an emerging field, high methodological heterogeneity was anticipated, and thus a meta-analysis was not planned. Eligible articles are grouped and synthesised according to outcome.

Quality assessment

Included studies were assessed for methodological quality using the Joanna Briggs Institute Checklist for Analytical Cross-Sectional Studies [34]. Quality assessment was undertaken independently by three authors (KR, RR, KF). Disagreements were resolved by discussion with a third rater (JT).

Results

The original search yielded 4368 articles. After removal of duplicates, 2865 articles remained. After title and abstracts were screened, 83 articles remained for full-text review. Of these, 34 were retained (see Fig. 1). During the hand search of reference lists, one additional study was identified, bringing the total number of included studies to 35.

Study and sample characteristics

Across the 35 studies, there was a cumulative sample of $N = 45,667$ (61% female, $M_{\text{age}} = 31.2$ years). All studies were cross-sectional, with the majority correlational in design (see Table 2). Only one study recruited a clinical sample [35], with an additional study utilising a sample of adults experiencing eco-anxiety [36]. The remaining studies recruited from the general population, with one study utilising a sample of pregnant women [37], ten studies conducted with University student samples, and three studies with young adults [38–40].

The studies were conducted across a range of regions including Europe (16 studies), North America (8 studies),

Australia/New Zealand (6 studies), the Philippines (1 study) and Turkey (1 study), with three conducted across multiple regions. Of the eight studies that reported ethnicity, all had a predominantly White sample [36, 40, 41, 44, 55, 63, 67, 69].

Significant gender differences in eco-anxiety were reported in five studies, with female-identifying participants reporting significantly higher eco-anxiety scores [41, 54, 59, 63, 68], although no gender differences were observed in one study [67]. Four studies indicated a relationship between younger age and higher eco-anxiety scores [41, 54, 59, 67].

Study quality

Study quality varied (Table 3). All studies assessed mental health outcomes using validated self-report tools, with none employing standardised clinical diagnostic assessments. Assessment and operationalisation of eco-anxiety varied across the studies. While twelve studies utilised validated eco-anxiety measures, three studies presented a validation of the exposure measure, a further 11 studies validated a language translation within the study, and two studies included or validated an adaptation of an existing measure within the study. The Climate Change Anxiety Scale (CCAS) was the most commonly used measure. Seven studies utilised non-validated assessments of eco-anxiety. Nearly half of studies did not clearly define the inclusion criteria for their samples, whereas confounding factors were identified in the majority of studies [29], but mitigation strategies were not always stated (12 studies).

Associations between eco-anxiety and mental health outcomes

All but three of the included 33 correlational studies found eco-anxiety to be significantly positively associated with mental health symptomatology, with small-to-large associations reported ($r = .14$ to 0.60). These relationships were consistently found across varying age ranges and nationalities, as well as in specific populations including young adults [39, 40], antenatal women [37], and a clinical sample [35]. By contrast, one study reported a negative association with eco-anxiety and stress, and found no significant relationships between eco-anxiety with other mental health outcomes [52], and one study reported no relationship with pathological worry [19]. An additional correlational study found no significant association with mental health outcomes after adjustment for covariates [53]. In a similar vein, most analyses were unadjusted, with two studies reporting that associations were largely no longer significant after adjustment for covariates of other worries or mental health symptoms [63, 66].

Relationships also varied by operationalisation of eco-anxiety. Where eco-anxiety was operationalised as

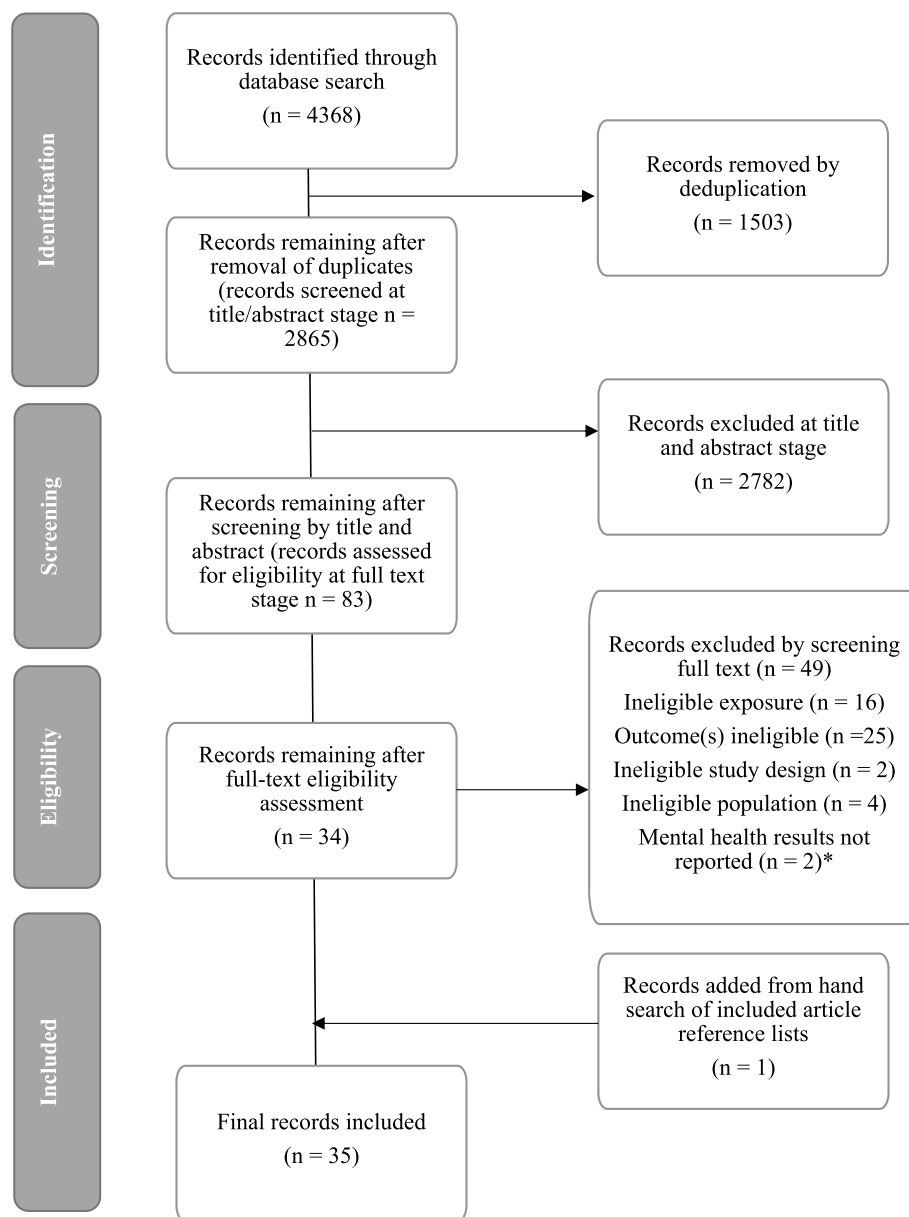


Fig. 1 PRISMA Flowchart showing number of studies at each stage of the review, reasons for exclusion and total number of studies retained

climate change-related worry, associations with mental health outcomes were more variable. Seven studies found small-to-moderate positive associations [38, 42, 45, 58, 63, 65, 66], whereas other studies showed no or minimal significance in adjusted analyses [19, 53, 63, 66], and one study found a negative association [52]. By contrast, 26 studies using measures of climate change-related anxiety consistently showed significant positive small-to-large associations, with only one study reporting no significant association in adjusted analyses [53]. Further, studies examining both climate change worry and climate

anxiety concurrently reported smaller associations for climate change worry compared with climate anxiety and studied mental health outcomes [36, 58, 65].

Psychological distress

All nine studies investigating symptoms of psychological distress found significant small-to-large ($r = .17$ to 0.70) positive associations [41, 43–45, 51, 54, 55, 68, 70]. Notably, some studies examined relationships using subscales of the exposure measures, with some variation reported between subscales. In one study

Table 2 Study characteristics and results relevant to the review question

Author, year	Country	Study design	Population & setting	Sample size and gender ^a %	Age range, mean (SD)	Exposure measure	Outcome measure	Results
Asgarizadeh et al. 2023 [36]	USA, CAN	Cross-sectional	Adults experiencing climate anxiety	N = 323 F = 65 M = 35 D = 0.1	19–87 M = 54.6 (16.7)	Climate change worry assessed by 10-items on 5-point Likert scale CCAS	Anxiety symptoms (GAD-7)	Moderate positive association between climate change worry and anxiety symptoms ($r = .40^{**}$) Large positive climate change anxiety and anxiety symptoms ($r = .60^{**}$)
Clayton & Karazsia 2020 [41]	USA	Cross-sectional	General population	N = 199 F = 32 M = 66 D = 1	18–84	CCAS	Psychological distress (PHQ-4)	Moderate and large positive associations between both climate anxiety subscales and distress ($r = .47^{***}$, 0.54^{***})
Curll et al. 2022 [42]	AUS	Cross-sectional	Representative Australian sample, stratified based on age, gender, and location	N = 3875 F = 50.7 M = 49 D = 0.3	18–89 M = 47 (17.5)	Climate worry – “How worried are you about global warming/climate change?” 4-point Likert scale (<i>not at all to very worried</i>) CCAS	Depression, anxiety, and stress symptoms (DASS-21)	Small positive associations between climate worry and depression ($r = .19^{**}$), anxiety ($r = .21^{**}$) and stress ($r = .24^{**}$)
Di Fabio & Svicher 2023 [43]	ITA	Cross-sectional	University Students	N = 411 F = 51 M = 49	M = 19.8	CCAS	Psychological distress (PHQ-4)	Small positive association between climate anxiety and distress ($r_s = 0.25^{***}$)
Feather & Williams 2022 [44]	AUS & NZ	Cross-sectional	General population	N = 771 F = 48 M = 50 D = 2	18–81 M = 33 (11.9)	CCAS-7	Psychological distress (PHQ-4)	Small positive association between total climate anxiety and distress ($r = .17^{***}$)
Gago & Sa 2021 [45]	PRT	Cross-sectional	University students	N = 106 F = 70 M = 30	18–25 M = 21.6 (1.8)	EWQ	Psychological distress (BSI-53)	Small positive association between environmental worry and distress ($r = .30^{**}$)
Gebhardt et al. 2023 [35]	GER	Cross-sectional	Psychosomatic clinic Outpatients	N = 89 F = 68 M = 31 D = 1	M = 37.6 (15.1)	GAD-7-C	Anxiety symptoms (GAD-7)	Moderate positive association between climate-related anxiety and anxiety symptoms ($r = .41^{***}$)

Table 2 (continued)

Author, year	Country	Study design	Population & setting	Sample size and gender ^a %	Age range, mean (SD)	Exposure measure	Outcome measure	Results
Hajek & Konig [46]	GER	Cross-sectional	General population	N = 3091 F = 50 M = 50 D = 0.2	18–74 M = 46.5 (15.3)	CCAS	Depression symptoms (PHQ-9), Anxiety symptoms (GAD-7)	Moderate positive association between climate anxiety with depression ($r = .30^{***}$), and anxiety ($r = .31^{***}$)
Heeren et al. 2023 [47]	BEL, FRA, SWZ	Cross-sectional	General population	N = 874 F = 51 M = 47 D = 1.5	18–81 M = 38.4 (14.1)	CCAS	Pathological worry (PSWQ)	Small positive associations between both climate anxiety subscales and worry, but worry explained the least variance in climate anxiety
Heinzel et al. 2023 [48]	GER	Cross-sectional	University Students and Staff	N = 486 F = 74 M = 24 D = 2	18–73 M = 29.4 (10.6)	HEAS	Depression, anxiety, and stress symptoms (DASS-21)	Small to moderate positive associations between the four eco-anxiety subscales with depression ($r = .27$ to 0.49^{***}), anxiety ($r = .23$ to 0.41^{***}) and stress ($r = .23$ to 0.43^{***})
Hepp et al. 2023 [49]	Various	Cross-sectional	General population from 10 countries (UK 42.5%; USA 34.2%; Canada 9.4%)	N = 374 F = 50 M = 47 D = 3	18–59 M = 30.7 (9.2)	CCAS	Depression symptoms (BDI-II), Anxiety symptoms (GAD-7)	Small positive association between climate anxiety subscales with anxiety ($r = .19$ to 0.20), negligible association with depression ($r = .02$)
Hogg et al. 2021 [50]	NZ	Cross-sectional	University Students	N = 365 F = 80 M/D = 20	M = 19.9 (3.59)	HEAS	Depression, anxiety, and stress symptoms (DASS-21)	Small to large positive associations between all three eco-anxiety subscales with depression ($r = .15$ to 0.37^{**}), anxiety ($r = .22$ to 0.46^{***}) and stress ($r = .22$ to 0.42^{***})

Table 2 (continued)

Author, year	Country	Study design	Population & setting	Sample size and gender ^a %	Age range, mean (SD)	Exposure measure	Outcome measure	Results
Innocenti et al. 2021 [51]	ITA	Cross-sectional	General population	N = 130 F = 67 M = 33	19–76 M = 32.8 (11.7)	CCAS	Anxiety symptoms (GAD-7, K10 anxiety subscale), depression symptoms (K-10 depression subscale), psychological distress (K10)	Positive association between climate anxiety cognitive-emotional impairment subscale and distress ($\beta = 0.44^{**}$), anxiety (GAD-7: $\beta = 0.89^{***}$, K10 anxiety subscale: $\beta = 0.14^{*}$) and depression ($\beta = 0.30^{*}$) Small association between climate anxiety functional impairment subscale and anxiety (GAD-7: $\beta = 1.35^{**}$) but not K10 anxiety subscale ($\beta = 0.23$), depression ($\beta = 0.37$) or distress ($\beta = 0.60$)
Innocenti et al. 2022 [52]	ITA	Cross-sectional	General population	N = 130 F = 65 M = 35	18–77 M = 35 (10.7)	CCWS	Depression, anxiety, and stress symptoms (DASS-21), Pathological worry (PSWQ)	Small negative correlation between climate worry and stress ($r = -.24^{*}$) but not depression, anxiety, or pathological worry
Innocenti et al. 2023 [53]	ITA	Cross-sectional	General adult population	N = 150 F = 53 M = 47	19–76 M = 35.5 (11.7)	CCAS CCWS HEAS	Depression, anxiety, and stress symptoms (DASS-21)	After adjustment for age and gender, no significant associations between any eco-anxiety measures with depression, anxiety or stress
Larionow et al. 2022 [54]	POL	Cross-sectional	General population	N = 603 F = 57 M = 41 D = 2	18–70 M = 25.3 (9.6)	CCAS	Depression and anxiety symptoms and psychological distress (PHQ-4)	Small positive associations between climate anxiety and distress ($r_s = 0.21^{*}$) and depressive symptoms ($r_s = 0.26^{**}$), but not anxiety symptoms ($r_s = 0.12$)

Table 2 (continued)

Author, year	Country	Study design	Population & setting	Sample size and gender ^a %	Age range, mean (SD)	Exposure measure	Outcome measure	Results
Lukacs et al. 2023 [55]	CAN	Serial Cross-sectional	General Population	N = 1553 F = 46 M = 50 D = 4	16–65	CCAS	Psychological Distress (K-6)	Large positive association between climate anxiety and distress ($r = .70^{***}$)
Lykins et al. 2024 [37]	AUS	Cross-sectional	Women in antenatal period	N = 103 F = 100	M = 30.8 (4.9)	CCAS	Worry symptoms (CWS), Depression symptoms (EPDS)	Moderate positive association between climate anxiety and depression anxiety and worry ($r = .37^{***}$), and worry ($r = .31^{***}$)
Mathé et al., 2023 [56]	CAN	Cross-sectional	University students	N = 275 F = 66 M = 32 D = 2	18–80 M = 31.2 (13.9)	HEAS	Anxiety (STAI-Y8), Depression, anxiety, and stress symptoms (DASS-21)	Small to moderate positive associations were found between all eco-anxiety subscales with anxiety (DASS-21: $r = .28$ to 0.37^{***} , STAI: $r = .29$ to 0.48^{***}), depression ($r = .26$ to 0.42^{***}), and stress ($r = .32$ to 0.46^{***})
Mouguiama-Daouda et al. 2022 [57]	FRA	Cross-sectional	General population	N = 305 F = 72 M = 27 D = 1	17–70 M = 30.8 (11.3)	CCAS	Anxiety symptoms (GAD-7), Depression symptoms (BDI-II)	Small positive association between climate anxiety with depression ($r = .30^{*}$), but not anxiety ($r = .02$)
Parmentier et al., 2024 [58]	FRA	Cross-sectional	General population	N = 431 F = 72 M = 27 D = 1	18–78 M = 37.6 (14.6)	CCAS EWS	Anxiety (STAI-Y)	Small positive associations between climate change anxiety subscales with anxiety ($r = .24$, $r = .26^{***}$) and between eco-worry with anxiety ($r = .18^{***}$)
Patrick et al. 2022 [59]	AUS	Cross-sectional	Representative sample, stratified by age, gender, location, and disadvantage	N = 5483 F = 58 M = 40 D = 2	18–75 M = 52.7 (20.0)	CCAS	Post-Traumatic Stress symptoms (PTSD-8)	5.76% of participants had both eco-anxiety and PTSD

Table 2 (continued)

Author, year	Country	Study design	Population & setting	Sample size and gender ^a %	Age range, mean (SD)	Exposure measure	Outcome measure	Results
Plohl et al. 2023 [38]	SLO	Cross-sectional	Young adults (18–24 years)	N = 442 F = 76 M = 22 D = 2	18–24 M = 21.6 (1.7)	CCAS CCWS	Anxiety, and stress symptoms (DASS-21)	Moderate positive associations between climate anxiety subscales with anxiety ($r = .30^{***}$) and stress ($r = .30$ to 0.31^{***}) Moderate positive associations between climate worry with anxiety ($r = .32^{***}$) and stress ($r = .36^{***}$) Moderate positive association between climate anxiety and distress ($r = .39^{***}$)
Reyes et al. 2023 [39]	PHL	Cross-sectional	Gen Z	N = 433 F = 67 M = 33	18–26 M = 20.4 (1.6)	CCAS	Psychological distress (MHI-38)	Moderate positive association between climate anxiety and distress ($r = .39^{***}$)
Schwaab et al. 2022 [60]	GER	Cross-sectional	Medical students	N = 203 F = 58 M = 42	M = 25.2 (3.7)	GAD-7-C	Anxiety symptoms (GAD-7),	Anxiety symptom scores were higher than the climate-related measures of anxiety ($d = 1.09^{***}$)
Schwartz et al. 2023 [61]	USA	Cross-sectional	University students	N = 284 F = 79 M = 19 D = 2	18–35 M = 23.1 (3.9)	CCAS	Anxiety symptoms (GAD-7), Depression symptoms (PHQ-8)	Small positive associations between climate anxiety subscales with anxiety ($r = .28^{***}$), and depression ($r = .24^{***}$) Positive associations between eco-anxiety and depression ($r = .28^{***}$), anxiety ($r = .30^{***}$), and stress ($r = .33^{***}$)
Stanley et al. 2021 [62]	AUS	Cross-sectional	General population	N = 3063 F = 56 M = 44 D = 0.3	M = 44 (17.5)	Eco-anxiety - extent climate change makes them feel anxious/afraid on a 0–100 scale (not at all to great deal)	Depression, anxiety, and stress symptoms (DASS-21)	

Table 2 (continued)

Author, year	Country	Study design	Population & setting	Sample size and gender ^a %	Age range, mean (SD)	Exposure measure	Outcome measure	Results
Stewart 2021 [63]	USA	Cross-sectional	University students	N = 417 F = 85 M/D = 15	18–37 M = 20.8 (1.9)	CCWS	Pathological worry (PSWQ), Depression, anxiety, and stress symptoms (DASS-21)	Small to medium positive associations between climate worry and depression ($r = .30^{***}$), anxiety ($r = .29^{***}$), stress ($r = .31^{***}$), and pathological worry score ($r = .17^*$) Only stress remained significant ($r = .12^*$) after adjustment for covariates
Thomson & Roach 2023 [64]	CAN	Cross-sectional	University students	N = 327 F = 72 M = 25 D = 3	18–68 M = 24.3 (9.2)	CCAS	Depression, anxiety, and stress symptoms (DASS-21)	Moderate to large positive associations between climate anxiety with depression ($r = .46^{**}$), anxiety ($r = .39^{***}$), stress ($r = .39^{***}$)
Türkarslan et al., 2023 [65]	TUR	Cross-sectional	General population	N = 605 F = 70 M = 30	18–68 M = 26.5 (8.3)	HEAS CCWS	Depression, anxiety, and stress symptoms (DASS-21)	Small positive associations between the eco-anxiety subscales with depression ($r = .24$ to 0.29^{***}), anxiety ($r = .29$ to 0.37^{***}), and stress ($r = .28$ to 0.32^{***}) Small positive associations between climate worry with depression ($r = .18^{***}$), anxiety ($r = .22^{***}$), and stress ($r = .22^{***}$)
Verplanken et al. 2020 [66]	Various	Cross-sectional	Adults residing in: US (59%), UK (22%), elsewhere (19%)	N = 266 F = 73 M = 25 D = 2	M = 26 (10.0)	Habit Index of Negative Thinking in relation to “Having worrying thoughts about global warming”	Pathological worry (PSWQ)	Positive association between climate-related worry and pathological worry ($r = .30$), not significant after adjustment for worry about personal issues

Table 2 (continued)

Author, year	Country	Study design	Population & setting	Sample size and gender ^y %	Age range, mean (SD)	Exposure measure	Outcome measure	Results
Verplanken et al. 2013 [19]	US and Europe	Cross-sectional	General population	N = 132 F = 59 M = 30 D = 1	18–67 M = 26 (10)	Habitual ecological worry	Pathological worry (PSWQ)	Ecological worry was not significantly associated with worry ($r = .05$)
Whitmarsh et al. 2022 [67]	UK	Cross-sectional	Broadly representative, by gender, age, and ethnicity	N = 1338 F = 53 M = 46.3 D = 0.7	18–85 M = 47.1	CCAS	Anxiety symptoms (GAD-7)	Small positive association between climate anxiety and anxiety ($r = .23^{***}$)
Wu et al. 2023 [40]	CAN	Cross-sectional	Adolescents aged 15–18 years	N = 2306 F = 46 M = 51 D = 3	15–18 M = 16.3 (0.5)	CCAS-Short form	Anxiety symptoms (GAD-2), Depression symptoms (PHQ-8)	Small correlations were found between climate anxiety with depression ($r_s = 0.14^{***}$) and anxiety ($r_s = 0.17^{***}$)
Wullenkord et al. 2021 [68]	GER	Cross-sectional	Adults stratified by age and gender	N = 1,011 F = 51 M/D = 49	18–69 M = 43.9 (14.0)	CCAS	Psychological distress (PHQ-4)	Small positive association between climate anxiety and distress ($r = .25^{**}$)

^y F denotes 'female', M denotes 'male', and D denotes diverse (other/non-binary/unspecified)

BDI-II Beck Depression Inventory, BSI Brief Symptom Inventory, CCAS Climate Change Anxiety Scale, CCWS Climate Change Worry Scale, CWS Cambridge Worry Scale, DASS-21 Depression Anxiety and Stress Scale, EPDS Edinburgh Postnatal Depression Scale, EWO Environmental Worry Questionnaire, EWS Eco-Worry Scale, GAD-2/GAD-7 Generalised Anxiety Disorder scale – 2 items/7 items, GAD-7 C Climate change-related anxiety, HEAS Hogg Eco-Anxiety Scale, K-6 and K-10 Kessler 6- or 10-item scale, MHI-38 The Mental Health Inventory, PHQ-4/-8/-9 Patient Health Questionnaire – 4 items/8 items/9 items, PSWQ Penn State Worry Questionnaire, PTSD-8 Posttraumatic Stress Disorder scale, STAI-Y State-Trait Anxiety Inventory, AUS Australia, BEL Belgium, CAN Canada, FRA France, GER Germany, ITA Italy, NZ New Zealand, PHL Philippines, POL Poland, PRT Portugal, SWZ Switzerland, UK United Kingdom, USA United States of America

* $p < .05$; ** $p < .01$; *** $p < .001$; β = linear regression coefficient; r = Pearson's correlation coefficient; r_s = Spearman's rho coefficient

Table 3 Study Quality assessed by JBI checklists for cross-sectional research

Study authors	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Asgarizadeh et al. 2023 [36]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Clayton & Karazsia 2020 [41]	No	Yes	Yes ^b	No	No	No	Yes	Yes
Curll et al. 2022 [42]	Yes	Yes	No	No	Yes	No	Yes	Yes
Di Fabio & Svicher 2023 [43]	No	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Feather & Williams 2022 [44]	Yes	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Gago & Sa 2021 [45]	Yes	Yes	No	No	No	No	Yes	Yes
Gebhardt et al., 2023 [35]	Yes	Yes	No	No	Yes	No	Yes	Yes
Hajek & Konig, 2023 [46]	No	Yes	Yes	No	Yes	Yes	Yes	Yes
Heeren et al. 2023 [47]	No	Yes	Yes	No	Yes	Yes	Yes	Yes
Heinzel et al. 2023 [48]	No	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Hepp et al. 2023 [49]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Hogg et al. 2021 [50]	No	Yes	Yes ^b	No	No	No	Yes	Yes
Innocenti et al. 2021 [51]	Yes	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Innocenti et al. 2022 [52]	Yes	Yes	Yes ^a	No	No	Yes	Yes	Yes
Innocenti et al. 2023 [53]	Yes	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Larionow et al. 2022 [54]	No	Yes	Yes ^a	No	No	No	Yes	Yes
Lukacs et al. 2023 [55]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Lykins et al. 2024 [37]	Yes	Yes	Yes	No	Yes	No	Yes	Yes
Mathé et al. 2023 [56]	No	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Mouguiama-Daouda et al. 2022 [57]	No	Yes	Yes ^a	No	Yes	No	Yes	Yes
Parmentier et al. 2024 [58]	No	Yes	Yes	No	Yes	Yes	Yes	Yes
Patrick et al. 2022 [59]	Yes	Yes	Yes	No	Yes	No	Yes	Yes
Plohl et al. 2023 [38]	Yes	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Reyes et al. 2023 [39]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Schwaab et al. 2022 [60]	Yes	Yes	No	No	Yes	Yes	Yes	Yes
Schwartz et al. 2023 [61]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Stanley et al. 2021 [62]	Yes	Yes	No	No	Yes	Yes	Yes	Yes
Stewart 2021 [63]	No	Yes	Yes ^b	No	Yes	No	Yes	Yes
Thomson & Roach 2023 [64]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Türkarslan et al. 2023 [65]	No	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Verplanken et al. 2020 [66]	No	Yes	No	No	Yes	No	Yes	Yes
Verplanken et al. 2013 [19]	No	Yes	No	No	No	No	Yes	Yes
Whitmarsh et al. 2022 [67]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Wu et al. 2023 [40]	No	Yes	Yes ^a	No	Yes	Yes	Yes	Yes
Wullenkord et al. 2021 [68]	No	Yes	Yes ^a	No	Yes	Yes	Yes	Yes

JBI, Joanna Briggs Institute (Moola et al., 2020)

The JBI cross sectional study checklist; Q1: Were the criteria for inclusion in the sample clearly defined?; Q2: Were the study subjects and the setting described in detail?; Q3: Was the exposure measured in a valid and reliable way?; Q4: Were objective, standard criteria used for measurement of the condition?; Q5: Were confounding factors identified?; Q6: Were strategies to deal with confounding factors stated?; Q7: Were the outcomes measured in a valid and reliable way? Q8: Was appropriate statistical analysis used?

^a Variation or language translation of a previously validated exposure measure used/validated

^b Exposure measure was validated within the study

[51], only the cognitive-emotional impairment subscale of the CCAS correlated with distress outcome, whereas both subscales (cognitive-emotional and functional impairment) were positively correlated with distress in another study ($r = .47$ and 0.54 , respectively) [41].

Depression symptoms

Eighteen studies investigated the association between eco-anxiety and symptoms of depression, with fifteen reporting a significant small-to-moderate ($r = .14$ to 0.49) positive association [37, 40, 42, 46, 48, 50, 51, 54, 56, 57, 62–65, 69], whereas three reported no significant

relationship [49, 52, 53]. In one study, only the cognitive-emotional impairment, but not functional impairment, subscale of the CCAS was related to depression [51], whereas all assessed subscales were positively related to depression in other studies using the CCAS [69], and the Hogg Eco-Anxiety Scale (HEAS) [48, 50, 56, 65].

Anxiety symptoms

Clinical anxiety symptoms were the most commonly examined, with twenty-three studies reporting on this relationship. Of the correlational studies, 18 showed significant small-to-moderate positive correlations between eco-anxiety and clinical anxiety symptoms ($r=.17$ to 0.60), including across all eco-anxiety measure subscales [35, 36, 38, 40, 42, 46, 48–51, 56, 58, 62–65, 67, 69]. Of note, the association with clinical anxiety symptoms was largest in the study with a sample experiencing eco-anxiety [36].

The pattern of positive association was largely noted across anxiety assessment, however, there was some variation in strength of association depending on the measure used. Measures assessing both state and trait anxiety simultaneously (e.g., State-Trait Anxiety Inventory) showed positive, albeit generally weaker correlations ($r=.24$ to $=0.26$) [56, 58], whereas studies utilising the GAD-7 more commonly reported stronger relationships than other measures. Notably, one study utilising both the GAD-7 and K10 anxiety subscale showed stronger correlations for the GAD-7 [51]. The size of correlations tended to vary widely when using the DASS-21, and studies utilising brief screening measures (e.g., PHQ 4-anxiety subscale and the GAD-2) found only small to negligible associations ($r=.12$ to 0.17).

Despite the general trend showing positive associations between eco-anxiety with clinical anxiety symptoms, three studies found the relationship to be non-significant [52, 54, 57], including one using a brief screener (PHQ 4-anxiety subscale). A further two studies showed eco-anxiety was not significant after adjustment for covariates [53, 63]. Additionally, in another study, climate-related anxiety (10%) was lower than overall anxiety symptoms (19%), suggesting that climate change anxiety did not account for anxiety-related psychological burden [60].

Stress symptoms

Nine studies, all utilising the DASS-21, reported significant small-to-moderate ($r=.22$ to 0.46) positive correlations between eco-anxiety with stress symptoms [38, 42, 48, 50, 56, 62–65]. The relationship was consistent for climate anxiety and climate change worry measures, as well as in samples from Europe, Oceania, and North America. Notably, one study found that, when controlling for other mental health outcomes, stress was the only assessed

mental health outcome that remained significantly associated with eco-anxiety [63]. Conversely, one study of the general Italian population found no significant association between any measure of eco-anxiety with stress after adjustment for covariates of age and gender [53], and another study utilising a similar Italian sample reported a small yet significant *negative* correlation between climate change worry and stress ($r=-.24$) [52].

Pathological worry

Five studies explored relationships between eco-anxiety with pathological trait worry. Two studies found small but significant ($r=.17$ to 0.30) positive relationships [63, 66], although these were no longer significant after adjustment. Another two studies showed no relationship between eco-anxiety and pathological trait worry [19, 52]. While the fifth study showed a small positive association between eco-anxiety and worry, worry was found to explain the least amount of variance in climate change anxiety [47]. By contrast, one study exploring state worry during the antenatal period found a moderate positive association with eco-anxiety [37].

Post-traumatic stress disorder

One study investigated the relationship between eco-anxiety and post-traumatic stress disorder (PTSD) symptoms. This study utilised cut-off scores to determine that 5.67% of participants were experiencing both eco-anxiety and threshold PTSD symptoms [59].

Discussion

The review showed that eco-anxiety was consistently associated with mental health outcomes including psychological distress, anxiety, depression, and stress symptoms. Only two studies, with similar samples and recruitment methods, showed no positive association between eco-anxiety and mental health symptoms. While relationships between eco-anxiety with PTSD and pathological worry were less consistent, overall, results indicate that *awareness* of climate change can impact functioning and poses psychiatric burden, in line with clinical observations [14]. While it has been noted that eco-anxiety is higher in younger samples [10, 17, 71], eco-anxiety is related to mental health outcomes across the lifespan, with similar results reported across the varied age ranges of samples in the included studies. Thus, irrespective of whether eco-anxiety represents a rational concern [18], results highlight the psychiatric burden associated with eco-anxiety and support calls from mental health care professionals for greater consideration of eco-anxiety in clinical practice, as well as in policy [16, 17].

While eco-anxiety was associated with mental health symptoms, there was evidence to suggest divergent

patterns of association with state vs. trait-based outcomes. Specifically, eco-anxiety was consistently associated with state-based outcomes [72] of depression, stress, and state-anxiety, suggesting that future research may benefit from using state measures such as the State Anxiety Inventory when exploring associations. State-anxiety is understood as representing an emotional state related to perceptions of apprehension or dread [73], which may be more closely associated with eco-anxiety. By contrast, eco-anxiety was inconsistently or weakly associated with pathological trait worry, where worry represents an inherent response tendency [73]. However, where state worry was assessed, a positive moderate association was observed [37]. Further, two studies that assessed state and trait anxiety symptoms together showed smaller associations with mental health outcomes than those that assessed only state-based outcomes. The findings pertaining to eco-anxiety are thus in line with evidence suggesting that trait variables account for less variance in emotional distress than state variables [74]. Together, the findings raise the possibility that eco-anxiety reflects distress in response to appraisals of the current situation, more so than being driven by persistent and pervasive characteristics of individuals (i.e., ‘worriers’).

Debate remains as to whether eco-anxiety represents a normal and rational response or should be understood as an anxiety disorder, or subtype of a disorder [75]. While individuals are likely to experience eco-anxiety across a range of thresholds, with many likely to be at subclinical levels, evidence also highlights that individuals can experience impairment in functioning due to eco-anxiety [10]. Therefore, there appears to be a proportion for whom eco-anxiety becomes impairing in day-to-day life, analogous to other psychiatric disorders such as major affective disorders. While climate change is a real and urgent threat, this would not preclude anxiety around climate change from being considered potentially pathological, with high levels of fear and anxiety a hallmark feature of anxiety disorders, irrespective of whether threats are ‘real’ or not [76]. This would also be in line with major diagnostic taxonomies for trauma and stressor related disorders, such as PTSD or reactive attachment disorder, where distress occurs in relation to ‘real’ events and threats see [76]. The intention is not to pathologise eco-anxiety per se, but simply to recognise that, for some individuals, it may reach levels that impair functioning. Accordingly, then, in order to best support these individuals, an understanding of nosology around eco-anxiety is critical in order to guide and inform treatment planning and clinical practice.

Eco-anxiety has been considered to have conceptual overlap with Generalised Anxiety Disorder (GAD), with

anticipatory worry representing a core feature of eco-anxiety [12, 75]. By contrast, the current findings regarding trait worry suggest some divergence from GAD, where high trait worry is commonly observed e.g., [77]. It remains plausible that eco-anxiety and GAD might align more closely through intolerance of uncertainty, or other attentional biases. Eco-anxiety has also been considered in line with PTSD [75], however, results here suggest a minimal relationship where eco-anxiety stems from vicarious experiences and not necessarily direct exposure. Whether eco-anxiety might best be conceptualised as a subtype of GAD, or if it better aligns with other anxiety disorders, such as phobias [75], or could best be understood as a distinct anxiety or stress-related disorder remains unclear. Across types of anxiety symptoms measured, results showed consistent associations with eco-anxiety. For example, small-to-large associations were found when GAD-like symptoms were assessed (i.e., the GAD-7), despite the limited overlap of eco-anxiety with trait worry, while small-to-moderate associations were also found when assessing symptoms more consistent with panic (i.e., DASS-21). Panic disorder is characterised by an impending sense of doom, which might conceptually align with the anticipatory element of eco-anxiety. Thus, nosological questions remain. Further, no included studies utilised objective diagnostic assessments of mental disorders, and only one study recruited a clinical sample, both of which may limit understandings of eco-anxiety classification. Thus, continued investigation is needed to ascertain whether eco-anxiety warrants conceptualisation as a disorder (or subtype), and if so, the most appropriate classification.

Consistent with previous reviews [7, 30], there was variability in conceptualisations of “eco-anxiety” and its operationalisation, which might partly explain the divergence in findings identified in this review. Further, fewer than half of the studies used a previously validated eco-anxiety tool. There remains an ongoing need to clarify definitions and ensure research coalesces around a consistent conceptualisation and validated assessment of eco-anxiety, minimising parallel investigations and jingle-jangle fallacies see [78]. Another limitation characterising the field in general is that few studies were designed to adjust for possible effect modifiers, confounds, or reverse-causality. Two studies showed no or minimal significant relationships between eco-anxiety with mental health outcomes after adjustment, thus better consideration of covariates including other mental health symptoms is needed in future analyses in order to better understand the nature of the relationship of eco-anxiety with mental health outcomes. Further, while gender and age differences in eco-anxiety were reported in some studies, few studies adjusted for age or gender.

There remains minimal comparison of patterns of association between eco-anxiety and mental health outcomes across gender or age groups, which may be valuable in future research. The majority of the literature is also limited to cross-sectional associations; thus, directionality and longer-term outcomes remain uncertain.

Further, the majority of included studies were conducted in high-income countries from Europe, North America, and Australia and New Zealand, with previous research suggesting differences in eco-anxiety across high-income vs. middle- and low-income countries [10]. It has also been suggested that eco-anxiety is a cultural construct [18], and varied understandings and experiences of climate change have differing impacts on the relationship with mental health [10, 12]. Greater global representation in eco-anxiety research is especially needed given that climate change is impacting different countries and regions at different rates [1], and some of the most impacted regions are largely missing from the current body of research. Impacts may also be greatest among Indigenous populations [17] and individuals and cultures with strong place-based attachments [79], further underscoring the need for diversity and representative populations in future research.

Limitations

While the focus was on eco-anxiety arising from the *vicarious* experience of ecological threats, any direct experience of climate change or extreme weather events was rarely assessed or utilised as a covariate. Therefore, the review cannot discount the impact of direct exposure on results. Additionally, other factors that might impact anxiety sensitivity, such as personality traits, or additional risk factors were not assessed or controlled for in the included studies and, thus, were not able to be considered in this review; this may prove a fruitful area of future research. In focusing on eco-anxiety, the review does not address mental health in relation to other related but distinct concepts such as eco-grief. Additionally, while the review focused on reporting associations between eco-anxiety and mental health outcomes, some age and gender related differences in eco-anxiety were observed. While meta-analysis was not performed in this review, a meta-analysis with meta-regression for age and gender may be a valuable inclusion in future reviews. Our review was restricted to peer-reviewed publications and did not encompass grey-literature, meaning that our review's findings may not have incorporated data from small unpublished studies that may include null findings. As a narrative synthesis without a formal test for publication bias, such as Egger's test, the extent to which publication bias might impact upon the reported results cannot be ascertained. In a newer and developing field, publication

bias may be amplified and thus results presented here may over estimate putative relationships. However, a recent meta-analysis that included both published and unpublished studies showed that publication status did not moderate the relationship between the CCAS with wellbeing [80]. Additionally, studies were restricted to English language, which might limit understanding of diverse cultures and contexts. Further, while a strength of the review was the restriction to studies using validated measures of mental health, this may have precluded studies in some cultures where validated mental health measures are less widely used or available, especially where qualitative approaches to understanding eco-anxiety and mental health have been used e.g., [81–83]. Thus, a meta-synthesis of qualitative studies may also be valuable to complement this review and inform our understanding of eco-anxiety across diverse populations or from a wider range of countries. The restriction to validated self-report mental health outcomes also may have excluded studies exploring potentially important cognitive processes (e.g. rumination) or elements of the Research Domain Criteria.

Implications for Mental Health Practice and Policy

Although some variations in findings and methodological limitations of the extant literature remain, which temper the conclusions that can be drawn, this systematic review highlights the association between eco-anxiety with mental health. Therefore, ongoing consideration of eco-anxiety in relation to assessment, intervention, and policy is needed. The review results also support the need to equip mental health care workers with skills and knowledge to address the impacts of climate change [84, 85]. The findings suggest the necessity for mental health professionals to integrate considerations of environmental factors into their clinical practice. Eco-anxiety might exacerbate symptoms [3], thus additional consideration of eco-anxiety as part of case formulations might be beneficial, especially when working with groups likely to be at higher risk. Enhanced assessment of eco-anxiety or related distress is thus warranted in clinical practice for consideration in case conceptualisation and treatment planning. Inclusion of nature-oriented questions at intake interview [86], might aid in assessing eco-anxiety. Development of robust clinical cut-offs for eco-anxiety measures may be beneficial to guide clinical practice.

While the review points to a benefit of explicitly considering eco-anxiety in treatment plans, currently, evidence guiding best practice for eco-anxiety remain scarce, with few interventions for eco-anxiety currently evaluated [21, 87]. To date, a range of approaches for addressing eco-anxiety have been suggested, such as incorporating eco-therapy and consideration of the human-nature relationship [86]. Upskilling individuals

with coping strategies [88], and use of mindfulness and cognitive-based therapies alongside psychoeducation and trauma-informed care [17], have also been proposed. A scoping review also indicated that fostering resilience, connection with nature, and grief-informed approaches might be valuable in treatment for eco-anxiety [87]. However, little is known regarding the effectiveness of any such eco-anxiety informed interventions and further high-quality research is needed to guide clinical practice. A lack of sufficient mental healthcare resourcing to cope with impacts of climate change has also been highlighted [17, 89], with increased resources likely to be needed moving into the future [17, 90]. A stronger focus on prevention might thus be critical to reduce psychiatric burden of climate change. Prevention aimed at building coping strategies and resilience, both individually and at the community level, may be beneficial [17, 91, 92].

However, individual therapy and group-based interventions around eco-anxiety cannot occur in isolation of broader policy and systemic changes. The high prevalence of eco-anxiety [10] - be it at clinical or subclinical levels - underscores the need for broader level changes related to climate change. Efforts targeting resilience and fostering climate resilient communities are needed to protect mental health [93]. Changes at environmental policy level are also pertinent given that worsening climate change will continue to detrimentally impact mental health [21] and further current health and social inequalities [94] if climate change is insufficiently addressed. The broader psychosocial impacts of climate change also need to be addressed at social policy levels [89, 95], rather than relying only on clinical approaches to eco-anxiety. A range of social determinants of health are likely to be amplified as climate change progresses, such as employment, housing, food security and social security—all of which require policy level consideration [17].

Conclusions

Our review highlights the significant positive relationship between eco-anxiety with a range of mental health outcomes, underscoring that climate change appears to be related to mental health even indirectly through overarching awareness. While eco-anxiety may not represent a pathology or its own disorder, it was nonetheless found to be related to distress and psychiatric impairment and, thus, warrants additional research attention to guide clinical practice and policy to support wellbeing and emotional resilience into the future.

Abbreviations

PTSD	Post-traumatic Stress Disorder
GAD	Generalised Anxiety Disorder
DASS-21	Depression, Anxiety and Stress Scale – 21 items
CCAS	Climate Change Anxiety Scale
HEAS	Hogg Eco-Anxiety Scale

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

SC reviewed eligibility screening, data extraction and quality assessment, synthesised data, and drafted the manuscript. RR conducted the literature review, eligibility screening, data extraction, synthesised data, and quality assessment and contributed to drafting the manuscript. KF conducted the literature review, eligibility screening, data extraction and quality assessment, synthesised data, and contributed to drafting the manuscript. KR conducted the literature review, eligibility screening and quality assessment. JT conducted the literature review, eligibility screening and quality assessment. PT contributed to methods and drafting the manuscript. AL conceptualised the review, reviewed eligibility screening, and contributed drafting of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. IPCC. Climate change 2022: impacts, adaptation, and vulnerability. C. U.; 2022.
2. Rocque RJ, Beaudoin C, Ndjaboue R, Cameron L, Poirier-Bergeron L, Poulin-Rheault R-A, et al. Health effects of climate change: an overview of systematic reviews. *BMJ Open*. 2021;11(6):e046333.
3. Berry HL, Bowen K, Kjellstrom T. Climate change and mental health: a causal pathways framework. *Int J Public Health*. 2010;55(2):123–32.
4. Bourque F, Willox AC. Climate change: the next challenge for public mental health? *Int Rev Psychiatry*. 2014;26(4):415–22.
5. Doherty TJ, Levy B, Patz J. Mental Health impacts. *Climate Change and Public Health*: Oxford University Press; 2015. p. 0.
6. Albrecht G, Sartore GM, Connor L, Higginbotham N, Freeman S, Kelly B, et al. Solastalgia: the distress caused by environmental change. *Australas Psychiatry*. 2007;15(Suppl 1):S95–8.
7. Coffey Y, Bhullar N, Durkin J, Islam MS, Usher K. Understanding eco-anxiety: a systematic scoping review of current literature and identified knowledge gaps. *J Clim Change Health*. 2021;3:100047.

8. Clayton S, Manning C, Krygsman K, Speiser M. Mental health and our changing climate: impacts, implications, and guidance. APA; 2017.
9. Pihkala P. Anxiety and the Ecological Crisis: an analysis of Eco-anxiety and Climate anxiety. *Sustainability*. 2020;12(19):7836.
10. Hickman C, Marks E, Pihkala P, Clayton S, Lewandowski RE, Mayall EE, et al. Climate anxiety in children and young people and their beliefs about government responses to climate change: a global survey. *Lancet Planet Health*. 2021;5(12):e863–73.
11. Ogunbode CA, Doran R, Hanss D, Ojala M, Salmela-Aro K, van den Broek KL et al. Climate anxiety, wellbeing and pro-environmental action: correlates of negative emotional responses to climate change in 32 countries. *J Environ Psychol*. 2022;84:101887.
12. Ogunbode CA, Pallesen S, Böhm G, Doran R, Bhullar N, Aquino S, et al. Negative emotions about climate change are related to insomnia symptoms and mental health: cross-sectional evidence from 25 countries. *Curr Psychol*. 2023;42(2):845–54.
13. Mento C, Damiani F, La Versa M, Cedro C, Muscatello MRA, Bruno A et al. Eco-anxiety: an Evolutionary line from psychology to psychopathology. *Med (Kaunas)*. 2023;59(12):2053.
14. Hickman C. We need to (find a way to) talk about ... eco-anxiety. *J Social Work Pract*. 2020;34(4):411–24.
15. Hickman C. Eco-Anxiety in Children and Young People - A Rational Response, Irreconcilable Despair, or Both? *Psychoanalytic Study of the Child*. 2024;77(1):356–68.
16. Burke SEL, Sanson AV, Van Hoorn J. The Psychological effects of Climate Change on children. *Curr Psychiatry Rep*. 2018;20(5):35.
17. Crandon TJ, Dey C, Scott JG, Thomas HJ, Ali S, Charlson FJ. The clinical implications of climate change for mental health. *Nat Hum Behav*. 2022;6(11):1474–81.
18. Clayton S. Climate anxiety: psychological responses to climate change. *J Anxiety Disord*. 2020;74:102263.
19. Verplanken B, Roy D. My worries are rational, Climate Change is not: habitual ecological worrying is an adaptive response. *PLoS ONE*. 2013;8(9):e74708.
20. McQueen A. The Wages of Fear? Toward Fearing Well About Climate Change In: *Philosophy and Climate Change*. Edited by: Mark Budolfson, Tristram McPherson, and David Plunkett, Oxford University Press (2021). <https://doi.org/10.1093/oso/9780198796282.003.0008>.
21. Heeren A, Asmundson GJG. Understanding climate anxiety: what decision-makers, health care providers, and the mental health community need to know to promote adaptive coping. *J Anxiety Disord*. 2023;93:102654.
22. Gianfredi V, Mazziotta F, Clerici G, Astorri E, Oliani F, Cappellina M, et al. Climate Change Perception and Mental Health. Results from a systematic review of the literature. *Eur J Investig Health Psychol Educ*. 2024;14(1):215–29.
23. Pitt C, Norris K, Pecl G. A Systematic Review of Climate Emotions and Mental Health in adults. *Global Environ Psychol*. 2023;1:e11405.
24. Hwang AR, Wang M, Khan H, Chagwedera DN, Grzenda A, Doty B, et al. Climate change and mental health research methods, gaps, and priorities: a scoping review. *Lancet Planet Health*. 2022;6(3):e281–91.
25. Cianconi P, Betrò S, Janiri L. The Impact of Climate Change on Mental Health: a systematic descriptive review. *Front Psychiatry*. 2020;11:74.
26. Charlson F, Ali S, Benmarhnia T, Pearl M, Massazza A, Augustinavicius J et al. Climate Change and Mental Health: a scoping review. *Int J Environ Res Public Health*. 2021;18(9):4486.
27. Burrows K, Denckla CA, Hahn J, Schiff JE, Okuzono SS, Randriamady H, et al. A systematic review of the effects of chronic, slow-onset climate change on mental health. *Nat Mental Health*. 2024;2(2):228–43.
28. Walinski A, Sander J, Gerlinger G, Clemens V, Meyer-Lindenberg A, Heinz A. The effects of Climate Change on Mental Health. *Dtsch Arztebl Int*. 2023;120(8):117–24.
29. Mondal S, Kar J, Bagchi E, Mukhopadhyay U. Eco-anxiety in adolescents: a Meta-analysis of the Impact of Climate Change on Mental Health. *Climate Crisis, social responses and sustainability: Socio-ecological study on global perspectives*. Springer; 2024. pp. 347–68.
30. Boluda-Verdú I, Senent-Valero M, Casas-Escolano M, Matijasevich A, Pastor-Valero M. Fear for the future: eco-anxiety and health implications, a systematic review. *J Environ Psychol*. 2022;84:101904.
31. White BP, Breakey S, Brown MJ, Smith JR, Tarbet A, Nicholas PK, et al. Mental Health impacts of Climate Change among vulnerable populations globally: an integrative review. *Ann Glob Health*. 2023;89(1):66.
32. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev*. 2015;4(1):1.
33. APA. Diagnostic and statistical manual of mental disorders 5th Edition. Washington, DC: APA; 2013.
34. Moola S, Munn Z, Tufanaru C, Aromataris E, Sears K, Sfetcu R, et al. Systematic reviews of etiology and risk. In: Aromataris E, Munn Z, editors. *JBI Manual for evidence synthesis*. Adelaide, Australia: JBI; 2020.
35. Gebhardt N, Schwaab L, Friederich HC, Nikendei C. The relationship of climate change awareness and psychopathology in persons with pre-existing mental health diagnoses. *Front Psychiatry*. 2023;14:1274523.
36. Asgarizadeh Z, Gifford R, Colborne L. Predicting climate change anxiety. *J Environ Psychol*. 2023;90:102087.
37. Lykins AD, Bonich M, Sundaraja C, Cosh S. Climate change anxiety positively predicts antenatal distress in expectant female parents. *J Anxiety Disord*. 2024;101:102801.
38. Plohl N, Mlakar I, Musil B, Smrke U. Measuring young individuals' responses to climate change: validation of the Slovenian versions of the climate anxiety scale and the climate change worry scale. *Front Psychol*. 2023;14:1297782.
39. Reyes MES, Carmen BPB, Luminarias MEP, Mangulabnan S, Ogunbode CA. An investigation into the relationship between climate change anxiety and mental health among Gen Z filipinos. *Curr Psychol*. 2023;42(9):7448–56.
40. Wu J, Long D, Hafez N, Maloney J, Lim Y, Samji H. Development and validation of a youth climate anxiety scale for the Youth Development Instrument survey. *Int J Ment Health Nurs*. 2023;32(6):1473–83.
41. Clayton S, Karaszia BT. Development and validation of a measure of climate change anxiety. *J Environ Psychol*. 2020;69:101434.
42. Curll SL, Stanley SK, Brown PM, O'Brien LV. Nature connectedness in the climate change context: implications for climate action and mental health. *Translational Issues Psychol Sci*. 2022;8:448–60.
43. Di Fabio A, Svicher A. The Eco-generativity Scale (EGS): a New Resource to protect the Environment and Promote Health. *Int J Environ Res Public Health*. 2023;20(15):6474.
44. Feather G, Williams M. The moderating effects of psychological flexibility and psychological inflexibility on the relationship between climate concern and climate-related distress. *J Context Behav Sci*. 2022;23:137–43.
45. Gago T, Sá I. Environmental worry and wellbeing in young adult university students. *Curr Res Environ Sustain*. 2021;3:100064.
46. Hajek A, König HH. Climate Anxiety and Mental Health in Germany. *Climate*. 2023;11(8):158.
47. Heeren A, Mouguiama-Daouda C, McNally RJ. A network approach to climate change anxiety and its key related features. *J Anxiety Disord*. 2023;93:102625.
48. Heinzel S, Tschorn M, Schulte-Hutner M, Schäfer F, Reese G, Pohle C, et al. Anxiety in response to the climate and environmental crises: validation of the Hogg Eco-anxiety Scale in Germany. *Front Psychol*. 2023;14:1239425.
49. Hepp J, Klein SA, Horsten LK, Urbild J, Lane SP. Introduction and behavioral validation of the climate change distress and impairment scale. *Sci Rep*. 2023;13(1):11272.
50. Hogg TL, Stanley SK, O'Brien LV, Wilson MS, Watsford CR. The Hogg Eco-anxiety Scale: Development and validation of a multidimensional scale. *Glob Environ Change*. 2021;71:102391.
51. Innocenti M, Santarelli G, Faggi V, Castellini G, Manelli I, Magrini G, et al. Psychometric properties of the Italian version of the climate change anxiety scale. *J Clim Change Health*. 2021;3:100080.
52. Innocenti M, Santarelli G, Faggi V, Ciabini L, Castellini G, Galassi F, et al. Psychometric properties of the Italian version of the climate change worry scale. *J Clim Change Health*. 2022;6:100140.
53. Innocenti M, Perilli A, Santarelli G, Carluccio N, Zjalic D, Maran DA et al. How does climate change worry influence the relationship between climate change anxiety and Eco-paralysis? A moderation study. *Climate*. 2023;11(9):190.
54. Larionow P, Soltys M, Izdebski P, Mudło-Głagolska K, Golonka J, Demski M, et al. Climate change anxiety Assessment: the Psychometric properties of the Polish Version of the climate anxiety scale. *Front Psychol*. 2022;13:870392.

55. Lukacs JN, Bratu A, Adams S, Logie C, Tok N, McCunn LJ et al. The concerned steward effect: exploring the relationship between climate anxiety, psychological distress, and self-reported climate related behavioural engagement. *J Environ Psychol.* 2023;90:102087.
56. Mathé M, Grisetto F, Gauvrit N, Roger C. Psychometric validation of the French Version of the Hogg Eco-anxiety Scale (HEAS-FR). *Canadian Journal of Behavioural Science-Revue Canadienne Des Sciences Du Comportement*; 2023.
57. Mouguiama-Daouda C, Blanchard MA, Coussement C, Heeren A. On the measurement of climate change anxiety: French validation of the climate anxiety scale. *Psychol Belg.* 2022;62(1):123–35.
58. Parmentier ML, Weiss K, Aroua A, Betry C, Rivière M, Navarro O. The influence of environmental crisis perception and trait anxiety on the level of eco-worry and climate anxiety. *J Anxiety Disord.* 2024;101:102799.
59. Patrick R, Snell T, Gunasiri H, Garad R, Meadows G, Enticott J. Prevalence and determinants of mental health related to climate change in Australia. *Aust N Z J Psychiatry.* 2022;57:48674221107872.
60. Schwaab L, Gebhardt N, Friederich HC, Nikendei C. Climate Change Related Depression, anxiety and stress symptoms perceived by medical students. *Int J Environ Res Public Health.* 2022;19(15):9142.
61. Schwartz SEO, Benoit L, Clayton S, et al. Climate change anxiety and mental health: Environmental activism as buffer. *Curr Psychol.* 2023;42:16708–21. <https://doi.org/10.1007/s12144-022-02735-6>.
62. Stanley SK, Hogg TL, Leviston Z, Walker I. From anger to action: Differential impacts of eco-anxiety, eco-depression, and eco-anger on climate action and wellbeing. *J Clim Change Health.* 2021;1:100003.
63. Stewart AE. Psychometric properties of the climate change worry scale. *Int J Environ Res Public Health.* 2021;18(2):494.
64. Thomson EE, Roach SP. The relationships among nature connectedness, climate anxiety, climate action, climate knowledge, and mental health. *Front Psychol.* 2023;14:1241400.
65. Türkarslan KK, Kozak ED, Yildirim JC. Psychometric properties of the Hogg Eco-anxiety Scale (HEAS-13) and the prediction of pro-environmental behavior. *J Environ Psychol.* 2023;92:102147.
66. Verplanken B, Marks E, Dobromir AI. On the nature of eco-anxiety: how constructive or unconstructive is habitual worry about global warming? *J Environ Psychol.* 2020;72:101528.
67. Whitmarsh L, Player L, Jiongco A, James M, Williams M, Marks E, et al. Climate anxiety: what predicts it and how is it related to climate action? *J Environ Psychol.* 2022;83:101866.
68. Wullenkord MC, Tröger J, Hamann KRS, Loy LS, Reese G. Anxiety and climate change: a validation of the climate anxiety scale in a german-speaking quota sample and an investigation of psychological correlates. *Clim Change.* 2021;168(3):20.
69. Schwartz SEO, Benoit L, Clayton S, Parnes MF, Swenson L, Lowe SR. Climate change anxiety and mental health: environmental activism as buffer. *Curr Psychol.* 2022;42:16708–21.
70. Reyes MES, Carmen BPB, Luminarias MEP, Mangulabnan S, Ogunbode CA. An investigation into the relationship between climate change anxiety and mental health among Gen Z filipinos. *Curr Psychol.* 2023;42(9):7448–56.
71. Stone K, Blinn N, Spencer R. Mental Health impacts of Climate Change on women: a scoping review. *Curr Environ Health Rep.* 2022;9(2):228–43.
72. Arvidsdotter T, Marklund B, Kylén S, Taft C, Ekman I. Understanding persons with psychological distress in primary health care. *Scand J Caring Sci.* 2016;30(4):687–94.
73. Spielberger CD. Theory and research on anxiety. *Anxiety Behav.* 1966;1(3):413–28.
74. Nordahl H, Ebrahimi OV, Hoffart A, Johnson SU. Trait Versus State predictors of emotional distress symptoms: the role of the Big-5 personality traits, Metacognitive Beliefs, and strategies. *J Nerv Ment Dis.* 2022;210(12):943–50.
75. Bourban M. Eco-anxiety and the responses of ecological citizenship and mindfulness. In: Jay Kassiola J, Luke TW, editors. *The Palgrave Handbook of Environmental Politics and Theory.* Cham: Springer International Publishing; 2023. pp. 65–88.
76. American Psychiatric Association. *Diagnostic and statistical manual of mental disorders* (5th ed., text rev.). USA: APA; 2022.
77. Hirsch CR, Mathews A, Lequertier B, Perman G, Hayes S. Characteristics of worry in generalized anxiety disorder. *J Behav Ther Exp Psychiatry.* 2013;44(4):388–95.
78. Gonzalez O, MacKinnon DP, Muniz FB. Extrinsic convergent validity evidence to prevent Jingle and Jangle Fallacies. *Multivar Behav Res.* 2021;56(1):3–19.
79. Cunsolo A, Harper SL, Minor K, Hayes K, Williams KG, Howard C. Ecological grief and anxiety: the start of a healthy response to climate change? *Lancet Planet Health.* 2020;4(7):e261–3.
80. Gago T, Sargissson RJ, Milfont TL. A meta-analysis on the relationship between climate anxiety and wellbeing. *J Environ Psychol.* 2024;94:102230.
81. Gibson KE, Barnett J, Haslam N, Kaplan I. The mental health impacts of climate change: findings from a Pacific Island Atoll nation. *J Anxiety Disord.* 2020;73:102237.
82. Gibson K, Haslam N, Kaplan I. Distressing encounters in the context of climate change: idioms of distress, determinants, and responses to distress in Tuvalu. *Transcult Psychiatry.* 2019;56(4):667–96.
83. Lykins AD, Cosh S, Nunn PD, Kumar R, Sundaraja C. Io, Keimami leqataka vakalevu na vei gauna mai muri (we are worried about the future generation): experiences of eco-grief in rural indigenous fijians. *Global Environ Psychol.* 2023;1:e11447.
84. Maughan D, Berry H, Davison P. What psychiatrists should know about environmental sustainability and what they should be doing about it. *Int Psychiatry.* 2014;11(2):27–30.
85. Maughan DL, Berry HL. Mind games: standing by while the world ignores climate change. *BJPsych Int.* 2015;12(2):29–30.
86. Hasbach P. Therapy in the Face of Climate Change. *Ecopsychology.* 2015;7(4):205–10.
87. Baudon P, Jachens L. A scoping review of interventions for the treatment of Eco-anxiety. *Int J Environ Res Public Health.* 2021;18:18.
88. Hayes K, Blashki G, Wiseman J, Burke S, Reifels L. Climate change and mental health: risks, impacts and priority actions. *Int J Ment Health Syst.* 2018;12:28.
89. Berry HL, Waite TD, Dear KBG, Capon AG, Murray V. The case for systems thinking about climate change and mental health. *Nat Clim Change.* 2018;8(4):282–90.
90. Every-Palmer S, McBride S, Berry H, Menkes DB. Climate change and psychiatry. *Australian New Z J Psychiatry.* 2016;50(1):16–8.
91. Berry HL, Hogan A, Owen J, Rickwood D, Fragar L. Climate change and farmers' mental health: risks and responses. *Asia Pac J Public Health.* 2011;23(2 Suppl):s119–32.
92. Mah AYJ, Chapman DA, Markowitz EM, Lickel B. Coping with climate change: three insights for research, intervention, and communication to promote adaptive coping to climate change. *J Anxiety Disord.* 2020;75:102282.
93. Department of Health and Aged Care. *National Health and Climate Strategy.* Canberra: Commonwealth of Australia; 2023.
94. Rao M, Powell RA. The climate crisis and the rise of eco-anxiety. *BMJ Opin.* 2021;6.
95. Doherty TJ, Clayton S. The psychological impacts of global climate change. *Am Psychol.* 2011;66:265–76.

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